

```
# =====
# STAGE 1 - CONTAINER 1
# DATASET OVERVIEW
# =====

print("DATASET OVERVIEW")
print("=" * 50)
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

df = pd.read_csv("/content/upi_transactions_2024.csv")

print("Dataset loaded successfully!")
print("Shape:", df.shape)
rows, columns = df.shape

print(f"Number of records : {rows:,}")
print(f"Number of columns : {columns}")

print("\nDataset memory usage:")
memory_mb = df.memory_usage(deep=True).sum() / (1024 ** 2)
print(f"{memory_mb:.2f} MB")

print("\nIndex information:")
print(f"Index type      : {type(df.index).__name__}")
print(f"Index start      : {df.index.min()}")
print(f"Index end        : {df.index.max()}")
```

```
DATASET OVERVIEW
=====
Dataset loaded successfully!
Shape: (250000, 17)
Number of records : 250,000
Number of columns : 17

Dataset memory usage:
182.17 MB

Index information:
Index type      : RangeIndex
Index start      : 0
Index end        : 249999
```

```
# =====
# STAGE 1 - CONTAINER 2
# COLUMN AND DATA-TYPE PROFILE
# =====

print("COLUMN AND DATA-TYPE PROFILE")
print("=" * 70)

profile = pd.DataFrame({
    "Column": df.columns,
    "Data Type": df.dtypes.astype(str).values,
    "Non-Null Count": df.notna().sum().values,
    "Unique Values": df.nunique(dropna=True).values
})

display(profile)
```

COLUMN AND DATA-TYPE PROFILE

=====

	Column	Data Type	Non-Null Count	Unique Values	
0	transaction id	object	250000	250000	
1	timestamp	object	250000	248610	
2	transaction type	object	250000	4	
3	merchant_category	object	250000	10	
4	amount (INR)	int64	250000	10355	
5	transaction_status	object	250000	2	
6	sender_age_group	object	250000	5	
7	receiver_age_group	object	250000	5	
8	sender_state	object	250000	10	
9	sender_bank	object	250000	8	
10	receiver_bank	object	250000	8	
11	device_type	object	250000	3	
12	network_type	object	250000	4	
13	fraud_flag	int64	250000	2	
14	hour_of_day	int64	250000	24	
15	day_of_week	object	250000	7	
16	is_weekend	int64	250000	2	

Next steps:

[Generate code with profile](#)

[New interactive sheet](#)

```
# =====
# STAGE 1 - CONTAINER 3
# RECORD INSPECTION
# =====

print("FIRST 5 RECORDS")
print("=" * 50)
display(df.head())

print("\nLAST 5 RECORDS")
print("=" * 50)
display(df.tail())

print("\n5 RANDOM RECORDS")
print("=" * 50)
display(df.sample(5, random_state=42))
```


FIRST 5 RECORDS

=====

transaction id	timestamp	transaction type	merchant_category	amount (INR)	transaction_status	sender_age_gro
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```
# =====
# STAGE 1 - CONTAINER 4
# DATA QUALITY INSPECTION
# =====
```

```
print("DATA QUALITY INSPECTION")
print("=" * 60)
```

```
# Missing values
missing = df.isnull().sum()
```

```
# Duplicate rows
duplicates = df.duplicated().sum()
```

```
# Total missing cells
total_missing = missing.sum()
```

```
print(f"Total missing cells : {total_missing:,}")
print(f"Duplicate rows      : {duplicates:,}")
```

```
print("\nMissing values by column:")
missing_table = pd.DataFrame({
    "Column": missing.index,
    "Missing Values": missing.values
})
```

```
display(missing_table)
```

```
# Explicit conclusion
print("\nDATA QUALITY RESULT")
print("=" * 60)
```

```
if total_missing == 0:
    print("✓ No missing values were found.")
```

```
if duplicates == 0:
    print("✓ No duplicate rows were found.")
```

15:23:07

5 RANDOM RECORDS

=====

	transaction id	timestamp	transaction type	merchant_category	amount (INR)	transaction_status	sender_age
38683	TXN0000038684	2024-05-12 15:12:19	P2M	Healthcare	988	SUCCESS	
64939	TXN0000064940	2024-01-23 11:35:10	P2P	Transport	57	SUCCESS	
3954	TXN0000003955	2024-07-15 05:05:01	P2P	Shopping	5872	FAILED	
120374	TXN0000120375	2024-12-26 16:42:44	P2M	Transport	134	SUCCESS	

DATA QUALITY INSPECTION2024-08-17 12:19:28TXN000172862P2PEntertainment325SUCCESS

Total missing cells : 0

Duplicate rows : 0

Missing values by column:

	Column	Missing Values	
0	transaction id	0	
1	timestamp	0	
2	transaction type	0	
3	merchant_category	0	
4	amount (INR)	0	
5	transaction_status	0	
6	sender_age_group	0	
7	receiver_age_group	0	
8	sender_state	0	
9	sender_bank	0	
10	receiver_bank	0	
11	device_type	0	
12	network_type	0	
13	fraud_flag	0	
14	hour_of_day	0	
15	day_of_week	0	
16	is_weekend	0	

DATA QUALITY RESULT

=====

✓ No missing values were found.

✓ No duplicate rows were found.

Next steps:

Generate code with missing_table

New interactive sheet

```
# =====
# STAGE 1 - CONTAINER 5
# INITIAL NUMERICAL PROFILE
# =====

print("INITIAL NUMERICAL PROFILE")
print("=" * 70)

numeric_columns = df.select_dtypes(
    include=np.number
).columns

print(f"Number of numerical columns: {len(numeric_columns)}")

print("\nNumerical columns:")
for i, column in enumerate(numeric_columns, start=1):
    print(f"{i}. {column}")

print("\nBasic numerical summary:")
display/
```

```
display(\n    df[numeric_columns].describe().T\n)
```

INITIAL NUMERICAL PROFILE

Number of numerical columns: 4

Numerical columns:

1. amount (INR)
2. fraud_flag
3. hour_of_day
4. is_weekend

Basic numerical summary:

	count	mean	std	min	25%	50%	75%	max	
amount (INR)	250000.0	1311.756036	1848.059224	10.0	288.0	629.0	1596.0	42099.0	
fraud_flag	250000.0	0.001920	0.043776	0.0	0.0	0.0	0.0	1.0	
hour_of_day	250000.0	14.681032	5.188304	0.0	11.0	15.0	19.0	23.0	
is_weekend	250000.0	0.285348	0.451581	0.0	0.0	0.0	1.0	1.0	

✓ Stage 1 Conclusion

The UPI Transactions 2024 dataset was successfully loaded and inspected.

The initial inspection established the dataset dimensions, column structure, data types, record samples, unique-value information and numerical-field profile.

The data-quality check identified no missing values and no duplicate rows.

These results establish the baseline structure and quality of the raw dataset. The dataset can therefore be passed to the next stage, where relevant records will be acquired and filtered according to the analysis requirements.

Double-click (or enter) to edit